Please check the examination details below before entering your candidate information

Candidate surname	Other names

Centre Number | Candidate Number

Pearson Edexcel Level 3 GCE

Time 2 hours

Paper reference **8MA0/01**

Mathematics

Advanced Subsidiary
PAPER 1: Pure Mathematics
Mock Set 2

You must have:
Mathematical Formulae and Statistical Tables (Green), calculator

Total Marks

Candidates may use any calculator allowed by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided – *there may be more space than you need*.
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Inexact answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 14 questions in this question paper. The total mark for this paper is 100.
- The marks for **each** question are shown in brackets – *use this as a guide as to how much time to spend on each question*.

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ▶

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Pearson

1. The line l_1 passes through the points (6, –5) and (2, 1)

(a) Find the equation of l_1 giving your answer in the form $y = mx + c$ where m and c are constants to be found.

(3)

The line l_2 is perpendicular to l_1 and passes through the origin.

(b) Find the coordinates of the point of intersection of l_1 with l_2

(4)

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Question 1 continued

(Total for Question 1 is 7 marks)

2.

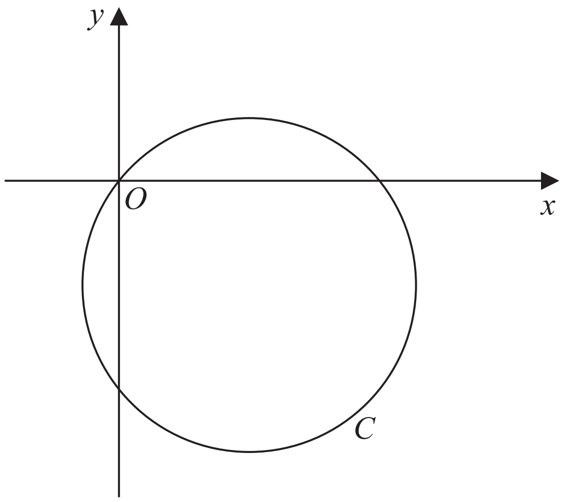

Figure 1

Figure 1 shows the circle C with equation

$$x^2 + y^2 - 12x + 10y = 0$$

(a) Find

(i) the coordinates of the centre of C

(ii) the exact radius of C

(3)

The straight line with equation $y = k$, where k is a positive constant, is a tangent to C at the point P

(b) Find the exact value of k

(1)

Given that C meets the y-axis at the origin O and at the point Q

(c) find the area of the triangle OPQ, making your method clear.

(3)

Question 2 continued

(Total for Question 2 is 7 marks)

3. **In this question you must show all stages of your working.**

Solutions relying entirely on calculator technology are not acceptable.

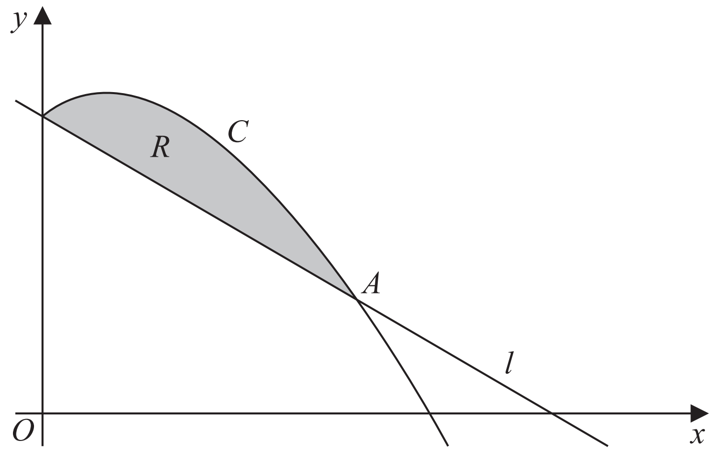

Figure 2

Figure 2 shows a sketch of part of the curve C with equation $y = 4x - 3x^{\frac{3}{2}} + 13$
and the line l with equation $y = 13 - 2x$

(a) Verify that the line and the curve intersect at the point $A(4, 5)$

(2)

The region R, shown shaded in Figure 2, is bounded by C and l

(b) Find the area of R

(4)

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Question 3 continued

(Total for Question 3 is 6 marks)

4. $$f(x) = x^4 - 2x^3 - 11x^2 + 12x + 36$$

(a) Use the factor theorem to show that $(x - 3)$ is a factor of $f(x)$

(2)

Given that $f(x) = (x - 3)^2(x + a)^2$, where a is a positive constant,

(b) deduce the value of a

(1)

(c) Sketch the curve with equation $y = f(x)$, indicating clearly on your sketch the coordinates of the points at which the curve crosses or meets the axes.

(4)

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Question 4 continued

(Total for Question 4 is 7 marks)

5. The straight line l has equation

$$2x - y + 6 = 0$$

The curve C has equation

$$y = 2x^2 + kx + 9$$

where k is a constant.

Find the range of possible values of k for which l meets C at 2 distinct points.

Give your answer in set notation.

(5)

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Question 5 continued

(Total for Question 5 is 5 marks)

6. Given

$$3^x = 3\sqrt{3}\left(9^{y+1}\right)$$

find an expression for y in terms of x in the form $y = ax + b$, where a and b are rational constants to be found.

(3)

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Question 6 continued

(Total for Question 6 is 3 marks)

7. Prove that, for all $x \in \mathbb{R}$,

$$4x^2 > 20x - 27$$

(3)

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Question 7 continued

(Total for Question 7 is 3 marks)

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8. (a) Find the first four terms, in ascending powers of x, of the binomial expansion of

$$(2 + 3x)^6$$

simplifying each term.

(4)

(b) Hence, without further calculation, write down the first four terms in the binomial expansion of

$$(2 - 3x)^6$$

(1)

(c) Hence, find the first two non-zero terms in the binomial expansion of

$$\left[(2 + 3x)^6 + (2 - 3x)^6\right]^2$$

(2)

(d) Find the term that is independent of x in the binomial expansion of

$$\left[(2 + ax)^n + (2 - ax)^n\right]^p$$

giving your answer in the form 2^y, where y is in terms of n and p

(2)

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Question 8 continued

Question 8 continued

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Question 8 continued

(Total for Question 8 is 9 marks)

9. **In this question you must show all stages of your working.**

Solutions relying entirely on calculator technology are not acceptable.

(a) Show that

$$5\cos\theta = 24\tan\theta$$

can be written as

$$5\sin^2\theta + 24\sin\theta - 5 = 0$$

(4)

(b) Hence solve, for $-180° \leqslant x < 540°$

$$5\cos x = 24\tan x$$

giving your answers to one decimal place.

(3)

(c) Deduce the number of solutions in the interval $-180° \leqslant x < 540°$ of the equation

$$5\cos(8x) = 24\tan(8x)$$

(1)

Question 9 continued

Question 9 continued

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Question 9 continued

(Total for Question 9 is 8 marks)

10.

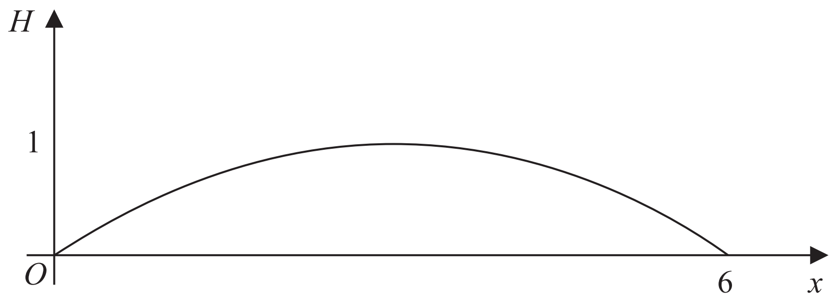

Figure 3

Figure 3 is a graph showing the trajectory of a snowboarder during a jump.

The vertical height, H metres, of the snowboarder above the ground has been plotted against the horizontal distance travelled, x metres, measured from where the snowboarder took off.

The snowboarder is modelled as a particle travelling in a vertical plane above horizontal ground.

The snowboarder

- reaches a maximum height of one metre above the ground
- lands on the ground at a point 6 metres from where the snowboarder took off

Given that H is modelled as a **quadratic** function in x

(a) find H in terms of x

(3)

The snowboarder passes over two vertical poles of height 0.5 metres.
The poles are in the same plane as the snowboarder's motion.

(b) Use the model to find the greatest distance between the two poles.

(4)

(c) Give one limitation of the model.

(1)

Question 10 continued

Question 10 continued

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Question 10 continued

(Total for Question 10 is 8 marks)

11. **In this question you must show all stages of your working.**

Solutions relying entirely on calculator technology are not acceptable.

Find any real values of x such that

$$2\log_6(x+3) = 2 - \log_6(4-x)$$

(7)

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Question 11 continued

(Total for Question 11 is 7 marks)

12. **In this question you must show all stages of your working.**

Solutions relying entirely on calculator technology are not acceptable.

The temperature, θ°C, of a cup of coffee as it cools is modelled by the equation

$$\theta = 22 + 64e^{-\frac{3}{32}t} \qquad t \geqslant 0$$

where t is the time in minutes since the cup of coffee was prepared.

(a) State the starting temperature of the coffee according to the model.

(1)

(b) Find the rate at which the coffee is cooling 10 minutes after it was prepared.

(3)

(c) Determine, according to the model, the length of time that the temperature of the coffee is between 40 °C and 60 °C. Give your answer to the nearest second.

(4)

Two hours after it was prepared, the temperature of the coffee is 20.8 °C.

(d) Evaluate the model in light of this information.

(1)

An insulated cup is designed to decrease the rate at which the coffee cools down.

(e) Explain how the model could be refined when this cup is used.

(1)

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Question 12 continued

Question 12 continued

Question 12 continued

(Total for Question 12 is 10 marks)

13.

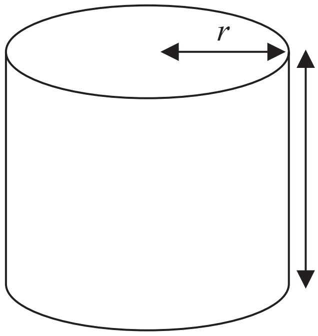

Figure 4

Figure 4 shows a closed cylindrical can of radius r cm and height h cm.

Given that the volume of the can is $400\,\text{cm}^3$

(a) show that the total surface area of the can, $A\,\text{cm}^2$, is given by

$$A = 2\pi r^2 + \frac{800}{r}$$

(4)

(b) Find the exact radius of the can for which A is a minimum.

(4)

(c) By finding $\frac{\mathrm{d}^2 A}{\mathrm{d}r^2}$, show that the radius found in part (b) gives the minimum value of A

(2)

(d) Calculate the minimum value of A

(2)

Question 13 continued

Question 13 continued

Question 13 continued

(Total for Question 13 is 12 marks)

14. In the triangle OAB

- $\overrightarrow{OA} = 3\mathbf{i} - 4\mathbf{j}$
- $\left|\overrightarrow{OB}\right| = 10$
- angle $AOB = 60°$

(a) Show that $\left|\overrightarrow{AB}\right| = 5\sqrt{3}$

(4)

Given further that $\overrightarrow{BA} = 4\sqrt{3}\,\mathbf{i} + p\,\mathbf{j}$, where p is a positive constant,

(b) find $\overrightarrow{OB}$, giving your answer using **i**, **j** notation.

(4)

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Question 14 continued

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Question 14 continued

(Total for Question 14 is 8 marks)

TOTAL FOR PAPER IS 100 MARKS